

Dead Signal

Product Requirements Document

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Version

Status

Last updated

Owner

Canon source

Companion docs

Purpose. This PRD defines *what* Dead Signal is, *who* it's for, and *what must be true* for it to ship. It is the product-and-requirements lens; the [GDD](#) covers *how it plays* and DESIGN.md covers *how it's built*. Narrative canon defers to STORY.md.

1. Overview

Dead Signal is a deterministic, hand-authored text-message survival mystery. You wake in the city of Harwick with no memory and a dying phone, guided by a stranger's texts toward a broadcast called Haven — and discover the evidence you find along the way is about you. The game is played entirely through a phone interface: reading messages, tapping choices, managing survival resources, and assembling a case file. It runs **100% offline** — no AI, no API keys, no network calls.

One-line positioning: *Lifeline's* intimacy meets *SOMA's* dread, in a survival game that turns the genre on the player.

2. Problem & opportunity

The opportunity. Text-message / interactive-fiction games have a passionate audience (*Lifeline*, *A Dark Room*, *Device 6*) but the format is underexplored for a *mystery with a real twist* and a *survival economy that carries dramatic weight*. Most texting games are either pure choose-your-own prose or real-time gimmicks. Few use the medium to make the player **complicit** in the story's horror.

The product bet. A short, dense, offline narrative game that:

1. Uses the phone frame as the entire aesthetic (low art cost, high immersion, mobile-native).
2. Hides a genuine philosophical horror twist (SOMA-style) under a survival crossing.
3. Is fully deterministic — canon-safe, replayable, and shippable anywhere with no backend.

Why now / why us. The core is already built and playable (see §9). The work is finishing to a professional standard: documentation, polish, accessibility, localization, and release.

3. Goals & success metrics

Product goals

- **G1 — Deliver a complete, coherent mystery.** The prologue cracks it; Phase 3 answers it; the finale lands. No plot holes (canon governed by STORY.md).
- **G2 — Make survival feel dramatic, not clerical.** Battery scarcity should read as tension, not bookkeeping.
- **G3 — Ship offline and deterministic.** Identical experience for every player, no backend.
- **G4 — Be broadly playable.** Mobile-first, installable, no-reflex input, accessible by design.
- **G5 — Reward replay.** Route variety + a completion meta-loop + two endings.

Success metrics (targets to instrument at release)

Metric	Target	Rationale
Prologue completion rate	≥ 60% of starters reach Haven	The crossing shouldn't lose most players
Phase 3 entry rate	≥ 50% of prologue-finishers continue	The handoff should pull players in
Full-arc completion (an ending)	≥ 30% of starters	Short game; completion should be strong
Both-endings / replay rate	≥ 15%	Validates the meta-loop
Crash / error rate	< 0.5% of sessions	Determinism should make this near-zero
Median prologue session	30–60 min	Design-intended length

Metrics are **local/opt-in only** (privacy pillar, §7.2). No telemetry ships without an explicit, documented opt-in.

4. Non-goals

- **Not** a live-service or content-treadmill game. It is a finite, authored experience.

- **Not** AI/LLM-driven. Live narration was prototyped and deliberately removed (see PLAN.md).
 - **Not** a combat game. Combat is a narrative device, intentionally shallow (GDD §7.3).
 - **Not** multiplayer, not social, not networked.
 - **Not** monetized in this build. Pricing/packageing is a separate future decision.
 - **Not** expanding *mechanics* pre-release (the STORY.md \$9 stop-doing list; GDD §14). **Note:** the sanctioned **Expansion v2** work (Echoes, region sub-stories, the unchosen, truth-by-assembly, the prologue second act; F-20–F-23) is *narrative depth, not new systems* — it is explicitly **in scope** as the answer to the story reading shallow/short, and adds no combat/survival mechanics.
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5. Target audience & personas

Primary audience. Narrative-game and interactive-fiction players; fans of *Lifeline*, *A Dark Room*, *SOMA*, *Her Story*, *80 Days*. Comfortable reading; want a story with teeth. Mobile-first.

Persona	Who	What they want	How we serve them
The IF reader ("Maya")	Plays <i>80 Days</i> , <i>Lifeline</i> on a phone in bursts	A tight, well-written mystery; meaningful choices	Phone-native, short sessions, a real twist
The dread-seeker ("Sam")	<i>SOMA</i> , <i>Silent Hill</i> , existential horror	Atmosphere and an idea that lingers	Restraint aesthetic; the upload horror; ambiguous endings
The completionist ("Devs")	Wants 100% and every ending	Systems to master, content to uncover	Case File meta-loop, 3 routes, 2 endings
The accessibility-first player	Needs no-reflex, readable, pausable play	To actually be able to play it	No-twitch design, pause-freezes-dialogue, muted-playable

Anti-persona. Players seeking action, fast reflexes, or systemic sandbox freedom — Dead Signal is deliberately not for them.

6. Platforms & technical constraints

- **Primary platform:** Web, delivered as an installable **PWA** (mobile-first, desktop-supported).
- **Tech stack:** React 18, Vite 5, Tone.js 15 (procedural audio). Single-file game component (src/DeadSignal.jsx). Full detail in DESIGN.md.
- **Offline:** must run with **no network** after load — no API keys, no fetches.
- **Persistence:** local only. Three save slots; a localStorage shim for standalone builds.
- **Distribution:** GitHub Pages workflow today; store/wrapper options (mobile app shells) are a future production decision.

- **Constraints to respect:** iOS Safari audio-unlock and standalone-PWA quirks (handled in audio.js / main.jsx); small-screen string budgets (a localization concern).
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7. Product requirements

Priority: **P0** = required to ship · **P1** = strongly wanted · **P2** = nice-to-have. Status: shipped · polish · planned.

7.1 Functional requirements

Narrative & flow

ID	Requirement	Priority	Status
F-1	Linear survival prologue: wake → contact → cross by one of 3 routes → shelter → Haven	P0	
F-2	Prologue ends on the call and auto- flows into Phase 3 (no dead-end completion screen)	P0	
F-3	Phase 3 hub-and- spoke investigation; 4 truth regions with progressive gating	P0	
F-4	Finale: final call + Accept/Refuse , two definitive endings, recorded per slot	P0	
F-5	Strict spoiler discipline: prologue cracks, Phase 3 answers (per STORY.md)	P0	
F-20	Echoes — recoverable fragments of the 142 minds (7-face cast); new ECHOES Case File category (Expansion v2)	P1	
F-21	Region sub-stories	P1	

	+ the <i>unchosen</i> thread; Phase-3 nodes ~2× (~35 → ~65) (Expansion v2)	
F-22	Truth-by-assembly — a region's truth resolves from its 2–3 supporting facts (Expansion v2)	P1
F-23	Prologue second act — route midpoints, the shelter as a scene, deeper Ellie; spoiler-safe (Expansion v2)	P2

Systems

ID	Requirement	Priority	Status
F-6	Survival economy: battery, charger reserve, food, water, HP, noise	P0	
F-7	Battery = life in P1–2; softens to exploration pressure in P3	P0	
F-8	Encounters via shared resolver with readable, computed risk tiers	P0	
F-9	Three routes with distinct resource identities + unique clue/fragment pools	P0	
F-10	Case File: board (memories/clues/people/locations) + journal (facts/questions)	P0	
F-11	Evolving Open Questions +	P1	

contradictions
(NEW/UPDATED
QUESTION cards)

F-12 Per-slot progression P1
accumulating
toward 100% across
runs

Presentation & shell

ID	Requirement	Priority	Status
F-13	Phone-native UI: message bubbles, HUD (signal/battery anchored), intent- coded choices	P0	
F-14	Human-paced Ellie texts; italic centered narrator; pause freezes dialogue	P0	
F-15	Title/menus, 3-slot save/load, Options (volume+mute), spoiler-safe Story page	P0	
F-16	Procedural audio incl. the story-gated Signal-distortion cue; fully playable muted	P1	
F-17	Installable PWA (manifest, icons, iOS meta) + marketing/demo shell	P1	
F-18	Accessibility options: text- speed/instant reveal, contrast/scaling, audio-cue visual equiv.	P1	
F-19	Localization: externalized strings	P2	

+ ≥1 non-English
language

7.2 Non-functional requirements

- **NFR-1 — Offline & deterministic (P0).** No network dependency post-load; identical outcomes given identical input. Zero API calls in production (verified; see PLAN.md M4).
 - **NFR-2 — Performance (P0).** Boots fast on mid-range mobile; smooth message/scroll on small screens; Tone.js lazy-loaded off the initial bundle.
 - **NFR-3 — Reliability (P0).** No broken build between milestones; saves load forward across schema changes (missing fields default). Crash rate < 0.5% (\$3).
 - **NFR-4 — Privacy (P0).** No data leaves the device by default. Any future telemetry is opt-in and documented.
 - **NFR-5 — Accessibility (P1).** WCAG 2.2 AA baseline; no reflex-gated beats; muted-playable. Full plan in ../accessibility/.
 - **NFR-6 — Localization-readiness (P2).** String architecture that supports externalization without refactoring the game logic. Plan in ../localization/.
 - **NFR-7 — Maintainability (P1).** Canon and systems documented; tuning knobs centralized and named.
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8. Release criteria (definition of done)

A build is release-ready when **all P0** functional and non-functional requirements are **and**:

1. A player can complete the **full arc** (start → an ending) on mobile and desktop with **zero console errors** and **zero network calls**.
2. Both endings are reachable and correctly recorded; saves survive a schema round-trip.
3. **No spoiler leaks** in any public surface (store copy, README, in-game Story page), verified against STORY.md \$4/\$7.
4. Narrative is **canon-consistent** — a full consistency pass against STORY.md finds no contradictions.
5. The PWA installs and runs offline on iOS Safari and Android Chrome; audio unlocks or safely no-ops.
6. Core-screen accessibility smoke test passes (readable, pausable, muted-playable, no-twitch).

Localization (F-19) and the full accessibility option set (F-18) are **P1/P2** — desired for a polished launch but not blocking a first release. They have their own delivery tracks.

9. Current state (what's built)

The playable spine is **complete**; current work is polish and consistency.

- **Prologue:** apartment wake, Ellie contact and identity setup, prep beats, route selection; full Phase-2 crossing on all three routes with atmosphere, encounters, memory fragments, and route clues; shelter; Haven approach; the empty-Haven finale with the impossible **143** record and the call.
- **Phase 3:** Haven hub + four truth spokes (Mercy General, Communications Array, City Hall, Research Annex) with progressive gating; the finale call; **Accept/Refuse** endings.
- **Systems:** survival economy with centralized tuning; shared encounter resolver with computed risk tiers; the declarative lead-queue exploration; the Case File (board + journal + evolving questions + contradictions), persisted per slot.
- **Shell:** title/menus, three save slots, Options, pause-freezes-dialogue, procedural audio, installable PWA, browser demo, GitHub Pages deploy.

See the [GDD §13](#) for the full content inventory and DESIGN.md for the technical state.

Planned next — Expansion v2 (design-locked; build: PLANNED). A narrative-depth pass, specified in the docs but **not yet in the game**: the **Echoes** (the 142 minds given voices, 7-face cast) as a new ECHOES Case File category; **region sub-stories** roughly doubling Phase-3 nodes; the **unchosen** moral thread (*why only 143*); **truth-by-assembly**; and a **prologue second act**. Requirements F-20–F-23; full canon in STORY.md “Changelog — Expansion v2”. This PRD tracks it as planned scope so current-state claims stay accurate (nothing here is shipped).

10. Milestones & roadmap (high-level)

Full schedule, exit criteria, and risk burndown belong to the **Production Plan** (../production/, planned). At a glance:

Phase	Focus	State
Core build	Prologue + Phase 3 + finale + endings	Done
Polish	Pacing, HUD readability, save/load, mobile, spoiler alignment	In progress
Docs & pipeline	GDD, PRD, DESIGN, then Production/Audio/Art/Loc/Access	In progress
Accessibility & localization	F-18 / F-19 tracks	Planned
Expansion v2 (narrative depth)	Echoes, region sub-stories, the unchosen, truth-by-assembly, prologue second act (F-20–F-23) — docs done; code planned	Planned

Release	Meet \$8 criteria; store/distribution decision	Planned
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11. Dependencies & risks

Dependencies

- STORY.md as the canon authority for every narrative requirement.
- Tone.js for audio; React/Vite for the app; GitHub Pages for delivery.
- iOS Safari behavior for PWA/audio (external, must be tested, not controlled).

Top risks (full register in the Production Plan)

Risk	Severity	Mitigation
Spoiler leak breaks the mystery	High	Spoiler discipline codified; public-copy review vs. vault
Narrative drift across docs/build	High	Single source of truth (STORY.md); standing consistency pass
Localization debt (hardcoded strings)	Med	Externalize early; loc is on the critical path (\$7.2 NFR-6)
iOS PWA/audio quirks	Med	Existing unlock/resume handling; keep under device test
Solo-dev bandwidth	Med	Phased docs; hard non-goals; polish over expansion

12. Open questions

- **Distribution & monetization.** Web-only free release, or wrap for mobile stores / itch / Steam? Pricing? (Out of scope for this build; must be decided before launch.)
 - **Localization scope.** Which languages ship first, and is loc a launch blocker or a fast-follow?
 - **Telemetry.** Do we want *any* opt-in metrics to validate \$3, or stay fully local?
 - **Phase 3 economy.** Should Phase 3 set its own low starting battery rather than inherit? (Design lever; see GDD \$15.)
 - **Expansion v2 held options.** Ship **U2** (a dedicated *citizen-shelter* region for the unchosen) and/or **X1** (endings shaded by what the player recovered)? Both are documented and out of the current pass; picking them up is an author's-call scope decision.
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13. Appendix

13.1 Related documents

- **Design:** [GDD](#) — how it plays.
- **Engineering:** [Technical Design](#) — how it's built.
- **Narrative canon:** STORY.md.
- **Docs hub:** ../README.md.
- **Historical:** PLAN.md — AI-removal + battery-economy roadmap (complete).

13.2 Change log

Version	Date	Notes
2.1	2026-07-06	Added Expansion v2 as planned scope: requirements F-20–F-23 (Echoes, region sub-stories + the unchosen, truth-by-assembly, prologue second act), §9 planned-next note, roadmap row, non-goals reconciliation, held options (U2/X1). All flagged build: PLANNED — nothing shipped.
2.0	2026-07-06	Full rewrite into a proper PRD: goals/metrics, personas, prioritized functional + non-functional requirements, release criteria, roadmap, risks. Supersedes the root prd.md build-status note.
1.0	2026-07	Original root prd.md — a build-status summary.

End of document.